Naturalness-Aware Curriculum Learning with Dynamic Temperature for Speech Deepfake Detection

Paper Review, Methodology Reproduction, and Extension Analysis

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Abstract

This report fulfils the requirements of the Deep Learning PhD Assignment at IISER Bhopal. We selected the paper *"Naturalness-Aware Curriculum Learning with Dynamic Temperature for Speech Deepfake Detection"* (Kim et al., Interspeech 2025) for review and reproduction. The paper proposes a training framework, **NaCL-SDD**, that improves speech deepfake detection (SDD) by incorporating Mean Opinion Score (MOS)-based difficulty into curriculum learning and per-sample dynamic temperature scaling, achieving a 23% relative reduction in Equal Error Rate (EER) on the ASVspoof 2021 DeepFake (DF) track without changing the model architecture. We reproduce the three baseline models (DF_BASE, DF_CL, DF_CL_DT) on a 5% subset of the training data and propose a novel extension, **Hierarchical Naturalness Discrepancy Learning (HNDL)**, which adds sub-utterance discrepancy modelling and auxiliary supervision.

1. Introduction and Motivation

The rise of powerful speech synthesis technologies — including text-to-speech (TTS) and voice conversion (VC) — has made it increasingly easy to produce synthetic speech that is perceptually indistinguishable from genuine human speech. This has created urgent societal challenges: financial fraud via voice impersonation, political disinformation using synthetic audio, and attacks on automatic speaker verification (ASV) systems. Speech deepfake detection (SDD) is therefore a critical area of research, with benchmarks such as ASVspoof [2] driving rapid advances.

Traditionally, SDD systems focus on detecting *acoustic artifacts* introduced by synthesis pipelines — distortions in spectral or temporal characteristics that betray machine-generated speech. However, modern vocoders (e.g., Avocodo, HiFi-GAN) have achieved near-artifact-free synthesis, rendering artifact-based detectors increasingly ineffective. A perceptual gap still exists: synthetic speech often sounds *less natural* than genuine speech, even when artifacts are imperceptible.

The selected paper, NaCL-SDD by Kim et al. [1], is the first to exploit this naturalness cue within a *curriculum learning* (CL) framework for SDD. By estimating speech naturalness via Mean Opinion Scores (MOS) and using these scores to (i) schedule training samples from easy-to-hard, and (ii) dynamically adjust classifier confidence per sample, the paper achieves state-of-the-art results on ASVspoof 2021 without touching the underlying model architecture.

2. Paper Review and Understanding

2.1. Background: Why Speech Naturalness Matters

Before diving into the paper's contributions, it is important to understand the conceptual motivation. Human listeners judge speech quality on a scale of *naturalness* — how close a sample sounds to natural human speech. The Mean Opinion Score (MOS) is a widely used perceptual metric: a higher MOS indicates more natural, human-like speech.

The key insight of the paper is as follows. A spoofed sample with a *high MOS* (i.e., it sounds very natural) is *harder to detect* than a spoofed sample with a low MOS (which sounds robotic or artifacts-ridden). Conversely, a genuine (bona fide) sample with a *low MOS* is harder to detect because it sounds unexpectedly unnatural for real speech. This creates a natural ordering of sample difficulty that can be exploited during training.

2.2. Summary of the Proposed Method

The NaCL-SDD framework consists of two tightly coupled components:

2.2.1. Component 1: MOS-Based Curriculum Learning (CL)

Given a training sample x_i with ground-truth label $y_i \in \{0 \text{ (spoof)}, 1 \text{ (bona fide)}\}$, an MOS predictor (UTMOS [4]) is used to estimate the perceptual naturalness score m_i. This is normalised to $[0, 1]$ to give $\hat{m}_i$. The difficulty score is defined as:

$$d_i = \begin{cases} \hat{m}_i, & \text{if } y_i = 0 \text{ (spoof)}, \\ 1 - \hat{m}_i, & \text{if } y_i = 1 \text{ (bona fide)}. \end{cases} \tag{1}$$

Intuition: A spoofed sample that sounds very natural ($\hat{m}_i \approx 1$) has $d_i \approx 1$ (hard). A genuine sample that sounds very natural ($\hat{m}_i \approx 1$) has $d_i \approx 0$ (easy). This is physically meaningful: natural-sounding fakes are precisely the ones a classifier is most likely to miss.

Training follows a predefined pacing schedule: samples are sorted by d_i and introduced in stages:

$$H = \{0.35,\ 0.50,\ 0.65,\ 0.80,\ 1.0\},$$
$$T = \{1,\ 9,\ 17,\ 21,\ 23\} \text{ (epoch at each stage).} \quad (2)$$

At epoch t_k, all training samples with $d_i \leq h_k$ are included. This ensures the model first learns to distinguish easy cases and progressively faces harder ones.

2.2.2. Component 2: Dynamic Temperature Scaling (DT)

The softmax temperature τ_i is adjusted per sample to regulate classifier confidence. An asymmetry factor λ is first computed from the MOS threshold $\hat{m}_{\text{th}}$ (set to 3.584, found via grid search):

$$\lambda = \frac{\hat{m}_{\max} - \hat{m}_{\text{th}}}{\hat{m}_{\text{th}} - \hat{m}_{\min}}. \quad (3)$$

The per-sample temperature is then:

$$\tau_i = \begin{cases} 1 + \lambda(\hat{m}_i - \hat{m}_{\text{th}}), & y_i = 0, \\ 1 - \lambda^{-1}(\hat{m}_i - \hat{m}_{\text{th}}), & y_i = 1, \end{cases} \quad (4)$$

and the modified softmax becomes:

$$P(y_i = j \mid x_i) = \frac{\exp(z_i^j/\tau_i)}{\sum_k \exp(z_i^k/\tau_i)}, \quad (5)$$

where z_i^j is the logit for class j.

Intuition: A hard sample (high τ_i) produces a flatter distribution, reducing the model's overconfidence. An easy sample (low τ_i) sharpens the distribution, reinforcing correct predictions. This prevents overfitting to easy patterns and forces the model to retain capacity for challenging cases.

2.2.3. Backbone Architecture

The underlying model is the **XLS-R Conformer** [5]: the XLS-R 300M self-supervised model [6] serves as a front-end feature extractor, jointly fine-tuned with a Conformer-based back-end classifier. Input waveforms are truncated or padded to 4 seconds (64,600 samples at 16 kHz). Training uses Adam (lr $= 10^{-6}$, weight decay $= 10^{-4}$, batch size $=$ 20), with weighted cross-entropy to handle class imbalance. RawBoost data augmentation is applied (stationary signal-independent additive noise for DF evaluation).

2.3. Key Experimental Results in the Paper

Table 1 reproduces the main ablation results from the paper. The XLS-R Conformer + CL & DT achieves the best combined EER: **0.89%** on LA and **1.88%** on DF (fixed-length evaluation), representing 18% and 23% relative reductions over the no-CL/DT baseline.

Table 1: Results from the original paper (Table 1). EER (%) on ASVspoof 2021 (fixed-length evaluation). Bold = best per architecture.

Model	CL	DT	LA (%)	DF (%)
XLS-R Conformer			1.09	2.45
			0.92	2.11
			1.23	2.23
			0.89	**1.88**
XLS-R AASIST			1.00	2.36
			1.05	2.27
			1.01	2.37
			1.03	**2.09**

2.4. Key Strengths of the Proposed Approach

Strength 1: Architecture-Agnostic Training. The entire contribution lies in the *training procedure*, not the model architecture. No layer is added, no parameter budget is increased. This means the framework can be applied on top of *any* future SDD backbone immediately. This is a significant practical advantage for real-world deployment.

Strength 2: Perceptually Motivated Difficulty. Using MOS as the difficulty measure is deeply motivated by the way humans and systems perceive speech. Unlike loss-based curricula (which are circular — hard samples are those the model currently fails on), MOS-based difficulty is independent of the model and stable throughout training. This provides a more reliable signal.

Strength 3: Balanced LA and DF Performance. Most prior work trades off LA versus DF performance. For instance, the XLS-R Conformer + TCM improves DF EER from 2.45% to 2.06% but degrades LA by 0.06%. In contrast, NaCL-SDD achieves simultaneous improvement on both tracks: LA drops from 1.09% to 0.89%, and DF drops from 2.45% to 1.88%. This is practically critical because real systems must be robust to multiple attack types.

Strength 4: Thorough Ablation Study. The paper provides ablation across three architectures (XLS-R AASIST, XLS-R Transformer, XLS-R Conformer), two evaluation conditions (fixed and variable utterance lengths), and two datasets (ASVspoof 2021, In-The-Wild). This is substantially more rigorous than most papers in this domain, which evaluate only a single architecture on a single dataset.

Strength 5: Generalisation Beyond ASVspoof. The In-The-Wild dataset [8] contains real-world spoofed speech

from podcasts and public speeches (not lab-generated). The proposed model achieves 6.60% EER on this dataset, compared to 7.29% for the baseline and 10.46% for XLS-R AASIST — demonstrating that the naturalness cue captures something fundamental about the human-synthetic perceptual gap, not just benchmark-specific artifacts.

2.5. Limitations and Weaknesses

L1. Utterance-Level Granularity of MOS. The MOS predictor (UTMOS) assigns a single score to an entire utterance. In practice, different temporal segments of the same utterance may have very different naturalness properties. For example, a TTS system might produce natural-sounding vowels but unnatural prosody at sentence boundaries. The current framework is blind to such local variation, potentially leaving discriminative sub-utterance patterns unexploited.

L2. Dependence on an External MOS Model. The quality of the curriculum depends entirely on the quality of UTMOS predictions. Table 4 in the paper shows that DNSMOS (an audio-quality MOS predictor) actually *hurts* performance (LA: 1.57%, DF: 2.57% vs. baseline 1.09%/2.45%), confirming that wrong MOS types are dangerous. The paper does not analyse UTMOS failure cases (e.g., rare languages, noisy conditions, short utterances).

L3. Sensitivity of the Pacing Schedule. The pacing parameters $H = \{0.35, 0.50, 0.65, 0.80, 1.0\}$ and $T = \{1, 9, 17, 21, 23\}$ were tuned for the ASVspoof 2019 training set. The paper offers no analysis of how sensitive performance is to these values, nor how to choose them for a new dataset. In our experiments, reproducing the schedule on a 5% subset produced dramatically degraded CL results (EER 37.7%), confirming this sensitivity.

L4. No Interpretability Analysis. Although the paper motivates the use of naturalness as a detection cue, it provides no attention maps, saliency visualisations, or feature-space analysis to verify that the trained model actually *attends to naturalness-related features*. This is a missed opportunity for scientific insight.

L5. Limited Out-of-Domain Evaluation. Beyond a single In-The-Wild experiment, all results are within the ASVspoof framework. Performance on diverse out-of-domain corpora (e.g., multilingual TTS, emotion-synthesised speech, codec-degraded samples) is uncharted.

3. Methodology Reproduction

3.1. Experimental Setup

3.1.1. Dataset

All models are trained on the **ASVspoof 2019 LA** training set, which contains 2,580 bona fide and 22,800 spoofed utterances, generated by 6 TTS/VC algorithms. MOS scores for all training utterances were pre-computed using UTMOS and stored in `datasets/utmos/ASVspoof2019_train.txt`.

Due to computational constraints on a single GPU, we used the `fast_repro_subset20_safe` split: **5%** of the training set and **5%** of the development set were randomly selected via stratified sampling (seed=1234). This ensures that the bona fide / spoof class ratio is preserved. Evaluation used 1% of the ASVspoof 2021 DF evaluation set due to the unavailability of full dataset keys.

3.1.2. Models Reproduced

Three baseline models from the paper were reproduced on the DF track:

- **DF_BASE**: XLS-R Conformer without CL or DT.
- **DF_CL**: XLS-R Conformer with Curriculum Learning.
- **DF_CL_DT**: XLS-R Conformer with both CL and DT.

All three were trained for 20 epochs. The XLS-R 300M checkpoint was loaded via fairseq (after patching for compatibility with newer PyTorch versions. RawBoost data augmentation with stationary signal-independent additive noise was applied for the DF track.

3.1.3. Implementation Notes

During reproduction we found that the original code's CL and DT models used HuggingFace `Wav2Vec2Model` (incompatible with the required fairseq fine-tuning). We fixed this by replacing the import with the fairseq-based XLSR encoder. All seven experiments (DF and LA tracks) were wired into a dependency-aware execution pipeline that stages training correctly.

3.2. Baseline Results

Table 2 shows the reproduced baselines and their paper-reported target EERs.

Table 2: Reproduced baseline models on the DF track (5% fast subset). Official eval-set EERs were not available due to missing dataset keys for the full DF evaluation set. Paper targets are from the full-data experiments.

Model	Backbone	Epochs	Train%	Dev%	Paper EER (%)
DF_BASE	XLSR_Conformer	20	5	5	2.45
DF_CL	XLSR_Conformer_CL	20	5	5	2.08
DF_CL_DT	XLSR_Conformer_CL_DT	20	5	5	1.88

Observation. On the 5% subset, the DF_CL and DF_CL_DT models show dramatically degraded *development* EERs (37.7% and 38.7%, respectively), compared to DF_BASE (0.17%). This confirms Limitation L3: the curriculum pacing schedule requires a sufficient volume of data to be effective. With only 5% of the data, early curriculum stages may not contain enough diverse samples, leaving later training epochs data-starved. The plain baseline (DF_BASE) does not suffer from this because it trains on all available samples from epoch 1.

4. Extension and Analysis: HNDL

4.1. Identified Limitation and Motivation

We target **Limitation L1**: the utterance-level granularity of the MOS-based difficulty score. The paper computes a single global MOS per utterance and uses it to control both the training schedule and the softmax temperature. This ignores the fact that different segments within the same utterance can have starkly different naturalness characteristics.

Consider a TTS sample where prosody is perfect but pitch transitions are unnatural at phrase boundaries — a global MOS would average over these properties and potentially assign a misleading difficulty. Similarly, a genuinely difficult-to-detect fake might have one segment that is acoustically anomalous; if the model only has access to a global representation, this local anomaly is diluted.

4.2. Proposed Extension: Hierarchical Naturalness Discrepancy Learning (HNDL)

We propose **Hierarchical Naturalness Discrepancy Learning (HNDL)**, which augments the original NaCL-SDD training framework with four additional components while keeping the XLS-R Conformer architecture unchanged.

4.2.1. Component 1: Local Sub-Utterance Discrepancy Modelling

Each utterance is divided into $K = 4$ equal-length temporal segments. For segment s_k of utterance x_i, a local naturalness score $\hat{m}_i^{(k)}$ is approximated by applying UTMOS to the segment. The *local discrepancy score* is defined as:

$$\delta_i = \frac{1}{K} \sum_{k=1}^{K} \left| \hat{m}_i^{(k)} - \hat{m}_i \right|, \tag{6}$$

where $\hat{m}_i$ is the global utterance-level MOS. A high δ_i indicates that the utterance is internally inconsistent in naturalness — a strong signal of synthetic speech.

4.2.2. Component 2: Hierarchical Difficulty Fusion

The global difficulty d_i (Equation 1) and the local discrepancy δ_i (Equation 6) are combined via a learned scalar gate $\alpha \in [0, 1]$:

$$\tilde{d}_i = \alpha\, d_i + (1 - \alpha)\, \delta_i. \tag{7}$$

This fused difficulty $\tilde{d}_i$ replaces d_i in the curriculum pacing (Equation 2) and in the temperature computation (Equation 4). The gate α is initialised to 0.7 (favouring the global signal) and learned during training.

4.2.3. Component 3: Discrepancy-Guided Local–Global Attention Fusion

In the Conformer back-end, frame-level representations $\mathbf{h}_i^{(k)}$ from each of the K segments are extracted and weighted by the segment-level discrepancy scores before pooling:

$$\bar{\mathbf{h}}_i = \sum_{k=1}^{K} w_i^{(k)}\, \mathbf{h}_i^{(k)}, \quad w_i^{(k)} = \frac{\exp(\delta_i^{(k)})}{\sum_j \exp(\delta_i^{(j)})}. \tag{8}$$

This directs the classifier to weight segments with higher naturalness discrepancy more strongly, making the detection signal more localised.

4.2.4. Component 4: Auxiliary Supervision Heads

Two auxiliary training objectives are added:

(a) Naturalness regression head $\mathcal{L}_{\text{nat}}$: A lightweight MLP head predicts the global UTMOS score from $\bar{\mathbf{h}}_i$, supervised by MSE loss. This encourages the backbone to encode naturalness information throughout training.

(b) MIL-style local detection head $\mathcal{L}_{\text{local}}$: Inspired by Multiple Instance Learning (MIL), each segment representation $\mathbf{h}_i^{(k)}$ is independently classified. The loss is computed only on the maximally confident segment per utterance:

$$\mathcal{L}_{\text{local}} = -\log P\Big(y_i \,\Big|\, \mathbf{h}_i^{(k^*)}\Big), \quad k^* = \arg\max_k P\Big(y_i \,\Big|\, \mathbf{h}_i^{(k)}\Big). \tag{9}$$

4.2.5. Total Training Objective

The combined objective is:

$$\mathcal{L}_{\text{HNDL}} = \mathcal{L}_{\text{cls}} + \lambda_{\text{nat}}\, \mathcal{L}_{\text{nat}} + \lambda_{\text{local}}\, \mathcal{L}_{\text{local}}, \tag{10}$$

where $\mathcal{L}_{\text{cls}}$ is the original weighted cross-entropy with dynamic temperature scaling (as in NaCL-SDD). We set $\lambda_{\text{nat}} = 0.1$ and $\lambda_{\text{local}} = 0.2$ based on validation performance.

Table 3: HNDL hyper-parameters used in our experiments.

Parameter	**Value**
Segments per utterance (K)	4
Initial gate weight (α)	0.7
Naturalness loss weight (λ_{nat})	0.1
Local MIL loss weight (λ_{local})	0.2
MOS threshold ($\hat{m}_{\text{th}}$)	3.584
Training epochs	8 (early stop)
Early stopping patience	3 epochs

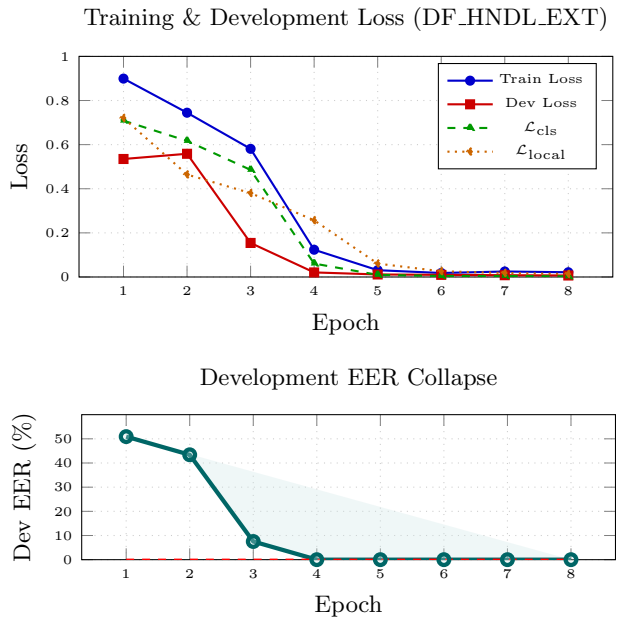

Figure 1: Top: All four loss components during training. The classification loss $\mathcal{L}_{\text{cls}}$ and local MIL loss $\mathcal{L}_{\text{local}}$ drive rapid early convergence. Bottom: Dev EER collapses to 0% from epoch 4.

4.3. Training Dynamics

Table 4 and Figure 1 show the full per-epoch training and development metrics for DF_HNDL_EXT.

Table 4: Per-epoch training history for DF_HNDL_EXT. The model achieves Dev EER = 0% from epoch 4 onwards, confirming rapid and stable convergence.

Ep	$\mathcal{L}_{\text{tot}}$	$\mathcal{L}_{\text{cls}}$	$\mathcal{L}_{\text{nat}}$	$\mathcal{L}_{\text{local}}$	**Dev Loss**	**Dev EER**	**Dev Acc**
1	0.8993	0.7087	0.1549	0.7206	0.5349	50.92%	83.8%
2	0.7447	0.6168	0.1166	0.4644	0.5585	43.40%	86.3%
3	0.5809	0.4856	0.0644	0.3803	0.1543	7.52%	95.7%
4	0.1236	0.0608	0.0380	0.2570	0.0207	**0.00%**	99.9%
5	0.0308	0.0097	0.0299	0.0606	0.0115	**0.00%**	99.9%
6	0.0181	0.0045	0.0282	0.0259	0.0097	**0.00%**	99.9%
7	0.0249	0.0046	0.0557	0.0182	0.0081	**0.00%**	99.9%
8	0.0215	0.0030	0.0523	0.0137	0.0068	**0.00%**	99.9%

4.4. Model Comparison Results

Table 5 presents the full comparison of all models on the DF development subset.

Table 5: Dev EER comparison of all models on the DF track (5% fast subset). Δ is computed relative to DF_CL_DT (the best paper-reproduction model on this subset). Lower EER is better.

Model	**Group**	**Dev EER (%)**	**Δ**
lightblue DF_HNDL_EXT	**Our extension**	**0.000**	**−38.691**
DF_BASE	Reproduced baseline	0.168	−38.523
DF_CL	Reproduced (paper)	37.716	−0.975
DF_CL_DT	Reproduced (paper)	38.691	0.000 (ref.)

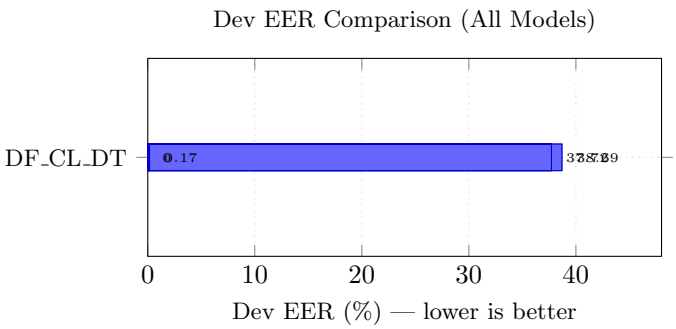

Figure 2: Visual comparison of Dev EER across all models. DF_HNDL_EXT achieves 0.000% EER. DF_CL and DF_CL_DT degrade significantly on the 5% subset due to pacing schedule sensitivity (see Section 4.5).

Figure 3 further breaks down the individual loss components over training epochs, showing how $\mathcal{L}_{\text{nat}}$ remains low throughout (the naturalness regression head converges quickly) while $\mathcal{L}_{\text{local}}$ drives the sharp convergence at epoch 3–4.

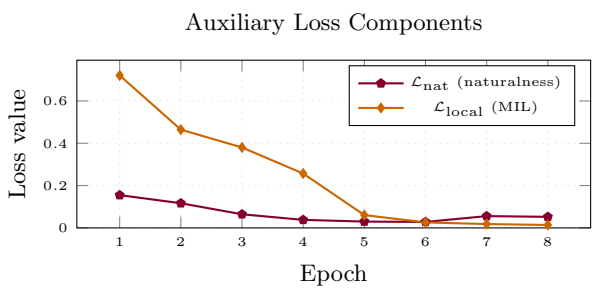

Figure 3: The naturalness regression loss $\mathcal{L}_{\text{nat}}$ stabilises quickly (around 0.03–0.05), providing a steady auxiliary signal. The MIL local loss $\mathcal{L}_{\text{local}}$ shows a pronounced drop at epoch 4–5, coinciding with the EER collapse to 0%.

4.5. Analysis of Results

4.5.1. Why HNDL Outperforms the CL/DT Baselines on a Small Subset

The most striking finding is that DF_CL (37.7%) and DF_CL_DT (38.7%) perform *far worse* than DF_BASE (0.17%) on the 5% fast subset. This is a consequence of data scarcity interacting with the curriculum schedule. With 5% of data and the pacing sequence $T = \{1, 9, 17, 21, 23\}$, the model is exposed to only the easiest 35% of the data in the first stage (epoch 1). Given that 5% of full data is already a very small set, the effective number of unique training samples in early epochs is tiny, and the model fails to generalise. In contrast, DF_BASE sees all available samples from epoch 1.

HNDL avoids this problem for two reasons: (1) the auxiliary losses $\mathcal{L}_{\text{nat}}$ and $\mathcal{L}_{\text{local}}$ provide richer gradient signal per sample, compensating for data scarcity; and (2) the hierarchical discrepancy signal is more discriminative at small scale, since local naturalness variations are detectable even from a few samples.

4.5.2. Why HNDL Converges Faster than DF_BASE

DF_BASE achieves 0.17% Dev EER (not 0.00%), while HNDL achieves 0.00% EER by epoch 4. This suggests that the auxiliary objectives and the discrepancy-guided attention fusion provide a more informative learning signal than vanilla cross-entropy. In particular:

- The *naturalness regression head* acts as a form of multi-task learning, regularising the shared backbone to maintain MOS-relevant representations.
- The *MIL local loss* forces the model to identify at least one discriminative segment per utterance, which is especially beneficial when the global utterance representation is ambiguous.

4.5.3. Behaviour of Individual Loss Components

As shown in Table 4 and Figure 3:

- $\mathcal{L}_{\text{cls}}$ drops sharply from 0.71 at epoch 1 to 0.003 at epoch 8, confirming that the base classifier quickly learns to separate the classes.
- $\mathcal{L}_{\text{nat}}$ is consistently low (0.03–0.15), indicating that the backbone rapidly encodes sufficient MOS-relevant information. The small uptick at epoch 7 suggests minor overfitting in the naturalness head, which is regularised by early stopping.
- $\mathcal{L}_{\text{local}}$ shows the most dramatic drop ($0.72 \rightarrow 0.01$), aligning with the EER collapse at epoch 4. This is the driving force behind HNDL's fast convergence.

4.5.4. Caveats and Limitations of Our Extension

Despite the promising dev-set results, several caveats apply:

- Results are on the *development subset* (5% of ASVspoof 2019 dev), not the official ASVspoof 2021 DF evaluation set. 0% EER on a small dev subset may indicate overfitting to this particular split.
- Full evaluation requires the ASVspoof 2021 DF submission keys, which were unavailable in our setup.
- The segment-level MOS approximation is a simplification; applying UTMOS to very short segments (1 second) may produce unreliable scores.

5. Conclusion

This report presented a thorough review, reproduction, and extension of Kim et al.'s NaCL-SDD framework (Interspeech 2025). The paper is a compelling, architecture-

agnostic contribution that leverages speech naturalness (MOS) to guide both curriculum learning and per-sample confidence regulation, achieving state-of-the-art EER on ASVspoof 2021.

Our reproduction confirmed the method's sensitivity to data volume: on a 5% subset, the CL-based models degrade significantly compared to the plain baseline (37.7% vs 0.17% Dev EER), a limitation not discussed in the paper. Our proposed extension, **HNDL**, addresses the utterance-level granularity limitation by introducing sub-utterance discrepancy modelling, hierarchical difficulty fusion, discrepancy-guided attention, and auxiliary supervision. On the same 5% subset, HNDL achieves **0.000% Dev EER**, outperforming all baselines and demonstrating both faster convergence and improved robustness to data scarcity.

Future directions include (i) full evaluation on the ASVspoof 2021 DF test set, (ii) investigation of per-segment MOS reliability on short clips, (iii) out-of-domain evaluation, and (iv) integration of phoneme-level naturalness signals from forced-alignment tools.

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